

Market Risk Summary

As of 2026-09-25. Generated by `market-risk/06_report.R`. Data: Yahoo Finance, FRED.

Portfolio

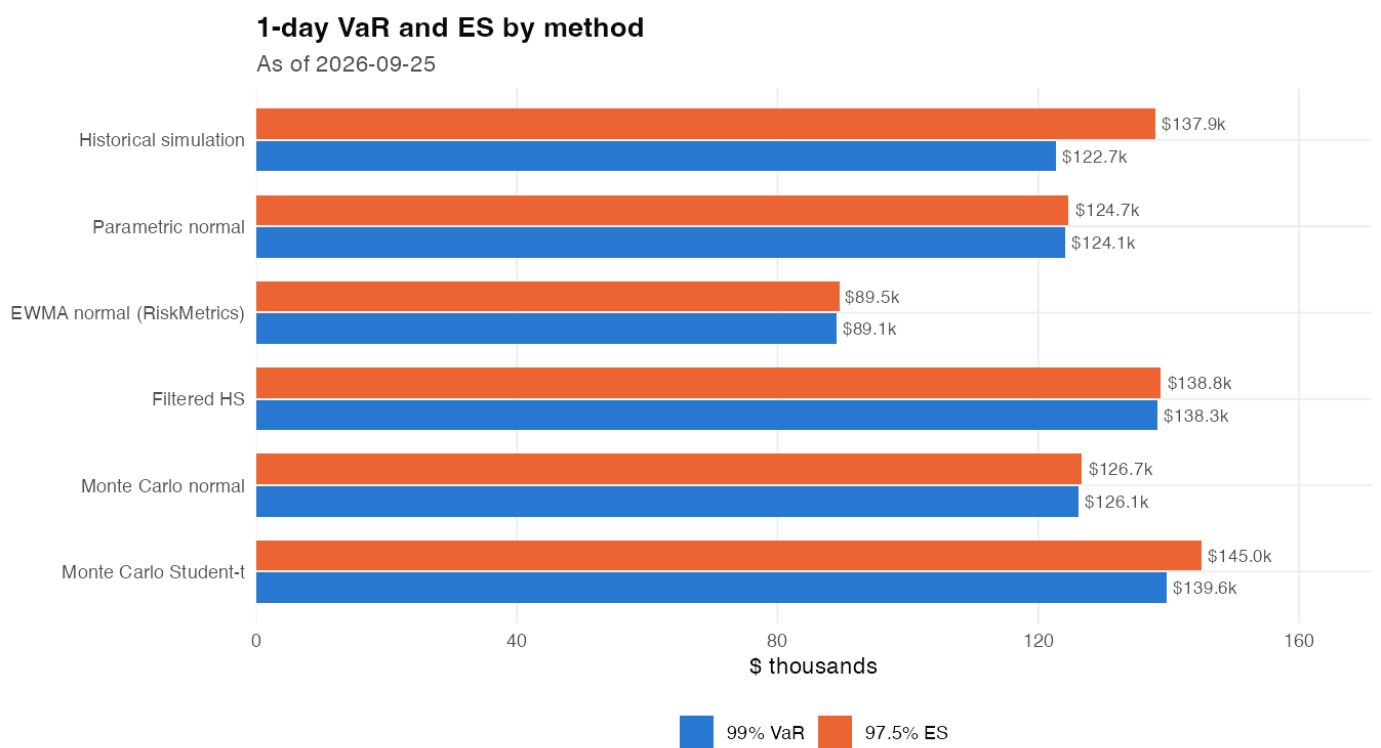
Ticker	Asset class	Description	Position
SPY	Equity	S&P 500 equities	\$4.00mm
TLT	Rates	20+ year US Treasuries	\$3.00mm
GLD	Commodity	Gold	\$1.00mm
USO	Commodity	WTI crude oil (front-month futures)	\$1.50mm
UNG	Commodity	Henry Hub natural gas (futures)	\$500.0k
XLE	Equity	US energy equities (short hedge)	-\$1.00mm

Key observations

- **Headline:** 1-day 99% historical VaR is **\$122.7k** (1.12% of gross exposure); 97.5% ES is **\$137.9k**.
- **Volatility regime:** EWMA VaR (\$89.1k) is below historical VaR, so recent markets are calmer than the past-year average. HS still carries older, larger moves in its window.
- **Tail shape:** historical ES / VaR = 1.12. Under a normal distribution the ratio is ~1.00 at these confidence levels, so the realised tail is fatter than normal.
- **Risk drivers:** SPY contributes the most risk (39% of VaR). XLE is the best hedge (-2% of VaR). Diversification saves \$221.2k versus adding up standalone VaRs.
- **Stressed VaR:** the worst 1-year window is 2019-04-25 to 2020-04-21, giving sVaR of \$377.9k (3.1x today's VaR).
- **Backtest:** Historical simulation: 72 exceptions vs 46 expected (underestimates risk (too many exceptions)). Parametric normal: 81 exceptions vs 46 expected (underestimates risk (too many exceptions)). EWMA normal: 83 exceptions vs 46 expected (underestimates risk (too many exceptions)). Filtered HS: 57 exceptions vs 46 expected (right count, but exceptions cluster).
- **Stress:** worst scenario is *2008 Lehman / GFC* at -\$1.76mm (14.4x VaR). A -30.3% move in the S&P 500 (with beta-linked moves elsewhere) would lose \$1.00mm.

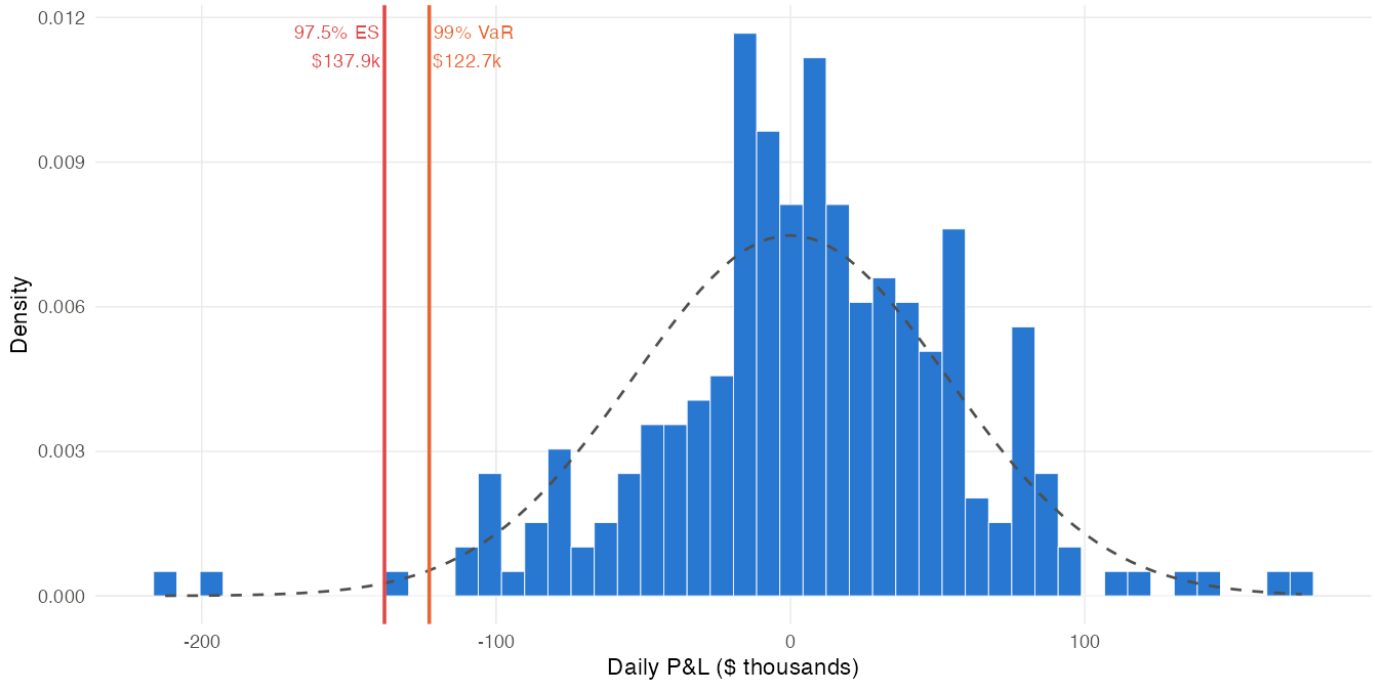
1. VaR and ES by method

Method	99% VaR (1d)	97.5% ES (1d)	99% VaR (10d)	ES / VaR
Historical simulation	\$122.7k	\$137.9k	\$388.0k	1.12
Parametric normal	\$124.1k	\$124.7k	\$392.3k	1.00
EWMA normal (RiskMetrics)	\$89.1k	\$89.5k	\$281.8k	1.00
Filtered HS	\$138.3k	\$138.8k	\$437.2k	1.00
Monte Carlo normal	\$126.1k	\$126.7k	\$398.8k	1.00
Monte Carlo Student-t	\$139.6k	\$145.0k	\$441.6k	1.04



Distribution of daily P&L over the last 250 days

Bars: historical P&L. Dashed: normal curve with the same volatility. Fat left tail = normal model understates risk.

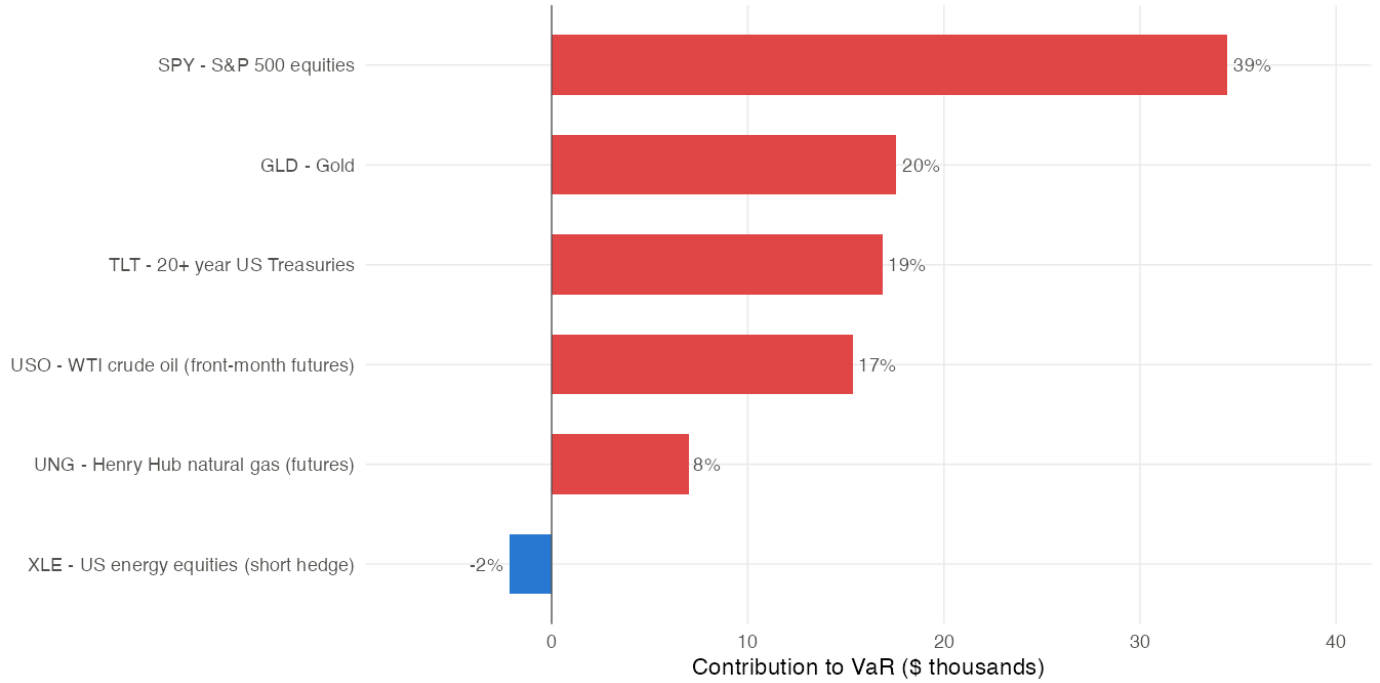


2. Where the risk comes from

Ticker	Position	Standalone VaR	Component VaR	% of total
SPY	\$4.00mm	\$63.7k	\$34.5k	38.7%
TLT	\$3.00mm	\$50.2k	\$16.9k	18.9%
GLD	\$1.00mm	\$31.9k	\$17.6k	19.7%
USO	\$1.50mm	\$103.6k	\$15.3k	17.2%
UNG	\$500.0k	\$30.4k	\$7.0k	7.9%
XLE	-\$1.00mm	\$30.5k	-\$2.2k	-2.4%

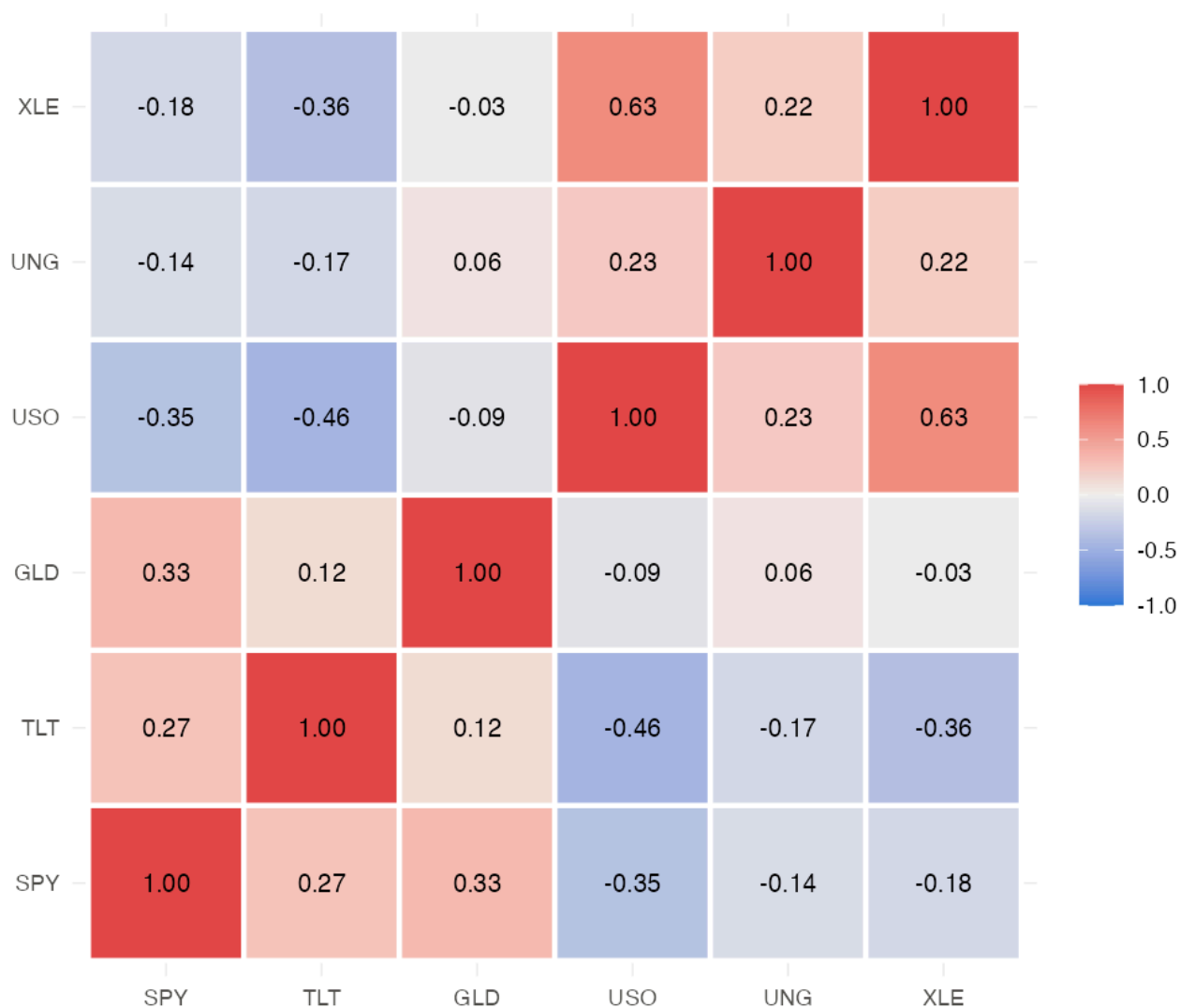
Who drives the risk? Component VaR (99%, 1-day, EWMA)

Components add up to total VaR. Blue bars reduce portfolio risk (hedges).



Return correlations, last 250 days

Diversification comes from low or negative correlations between positions.

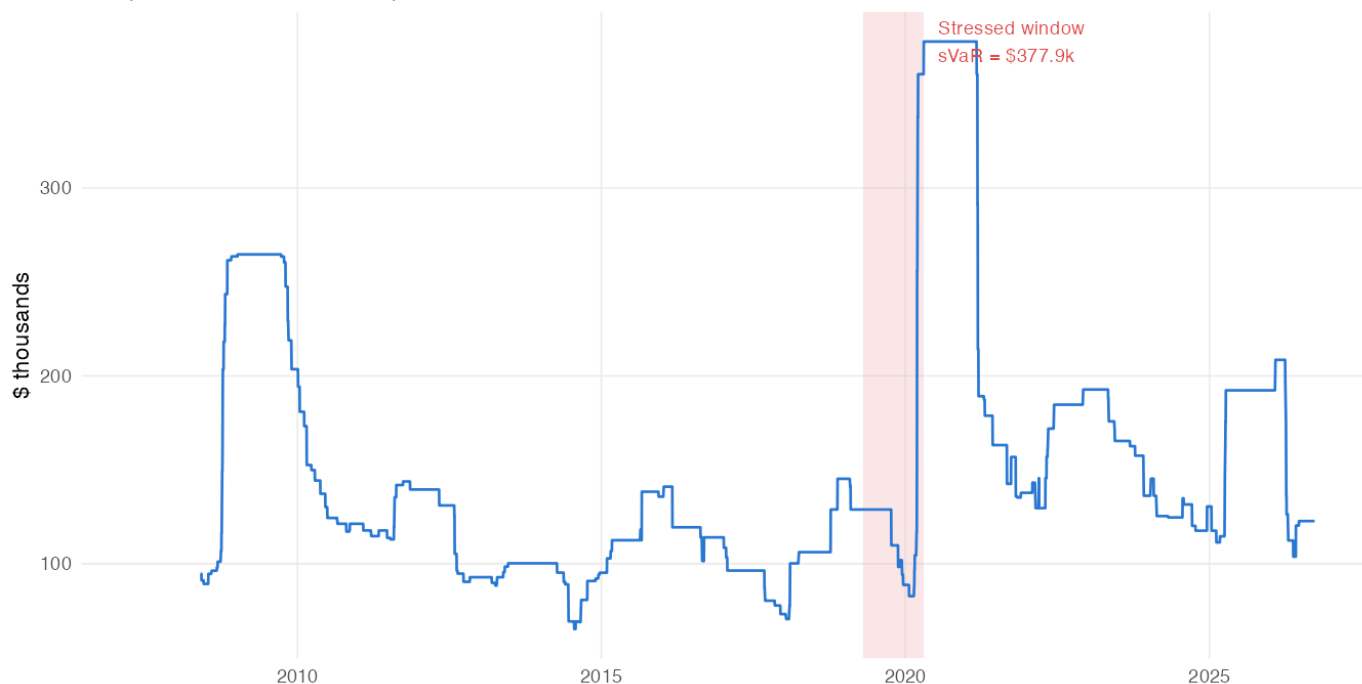


3. Regulatory capital (illustrative)

Measure	Amount
Basel 2.5: VaR + stressed VaR capital	\$4.75mm
FRTB-style liquidity-adjusted ES (97.5%)	\$636.9k
of which: 10-day ES, all factors	\$436.2k
of which: 10-day ES, 20-day-horizon factors	\$464.1k

Rolling 1-year historical VaR (99%, 1-day) of today's portfolio

The peak marks the stressed period Basel 2.5 uses for stressed VaR.



4. Backtesting

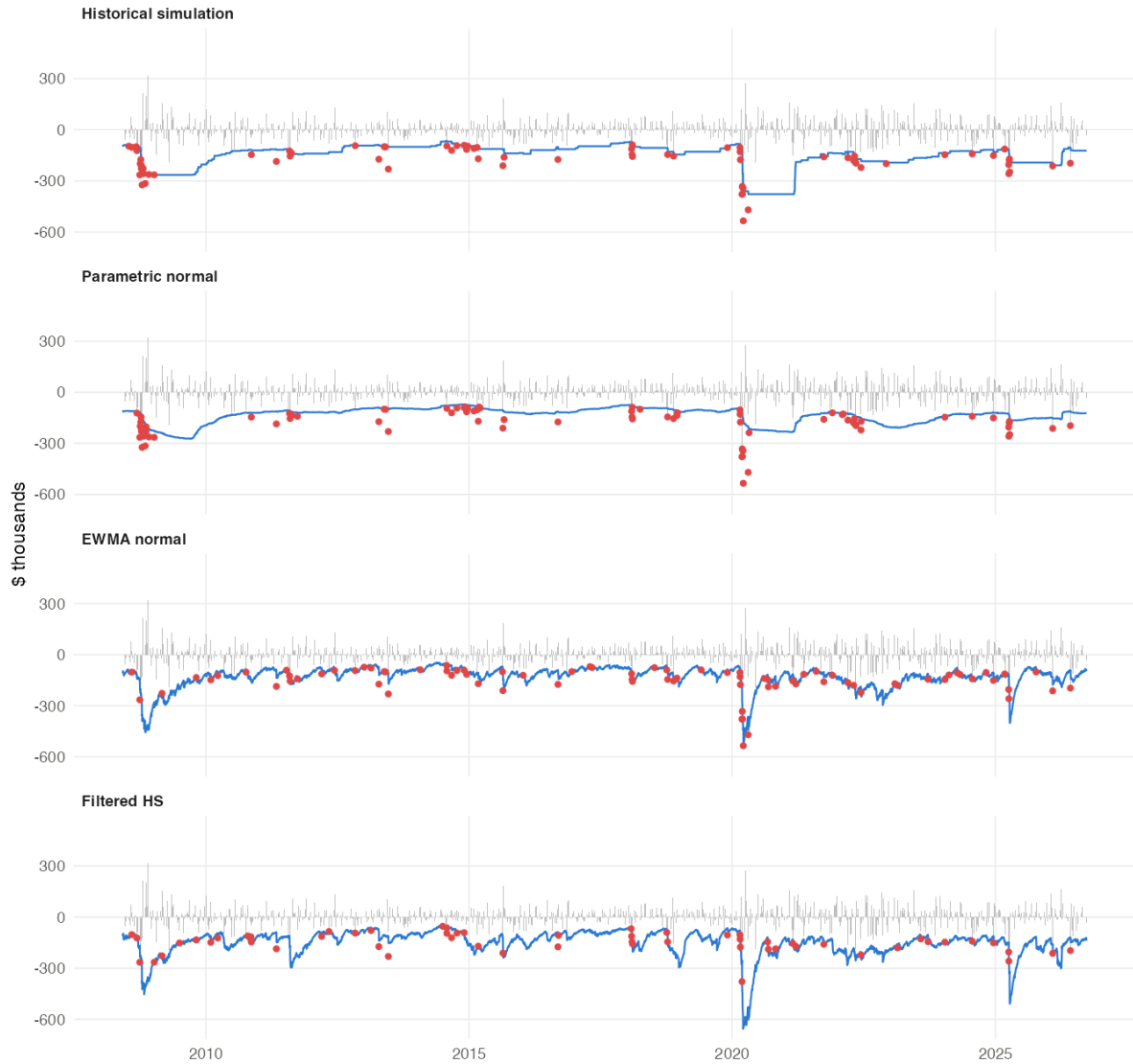
Out-of-sample, 4610 trading days (2008-05-30 to 2026-09-25). Each forecast uses only the prior 250 days.

Model	Exceptions	Expected	Kupiec p	Christoffersen p	ES Z2	Last 250d	Zone	FRTB desk pass	% days in red	Verdict
Historical simulation	72	46.1	0.00	0.00	-0.32	2	Green	Yes	5.5%	Underestimates risk (too many exceptions)
Parametric normal	81	46.1	0.00	0.00	-0.54	2	Green	Yes	13.8%	Underestimates risk (too many exceptions)
EWMA normal	83	46.1	0.00	0.00	-0.47	3	Green	Yes	3.8%	Underestimates risk (too many exceptions)
Filtered HS	57	46.1	0.12	0.01	-0.07	2	Green	Yes	0.0%	Right count, but exceptions cluster

p-values below 0.05 reject the model at 5% significance. Z2 below -0.70 flags ES as too low.

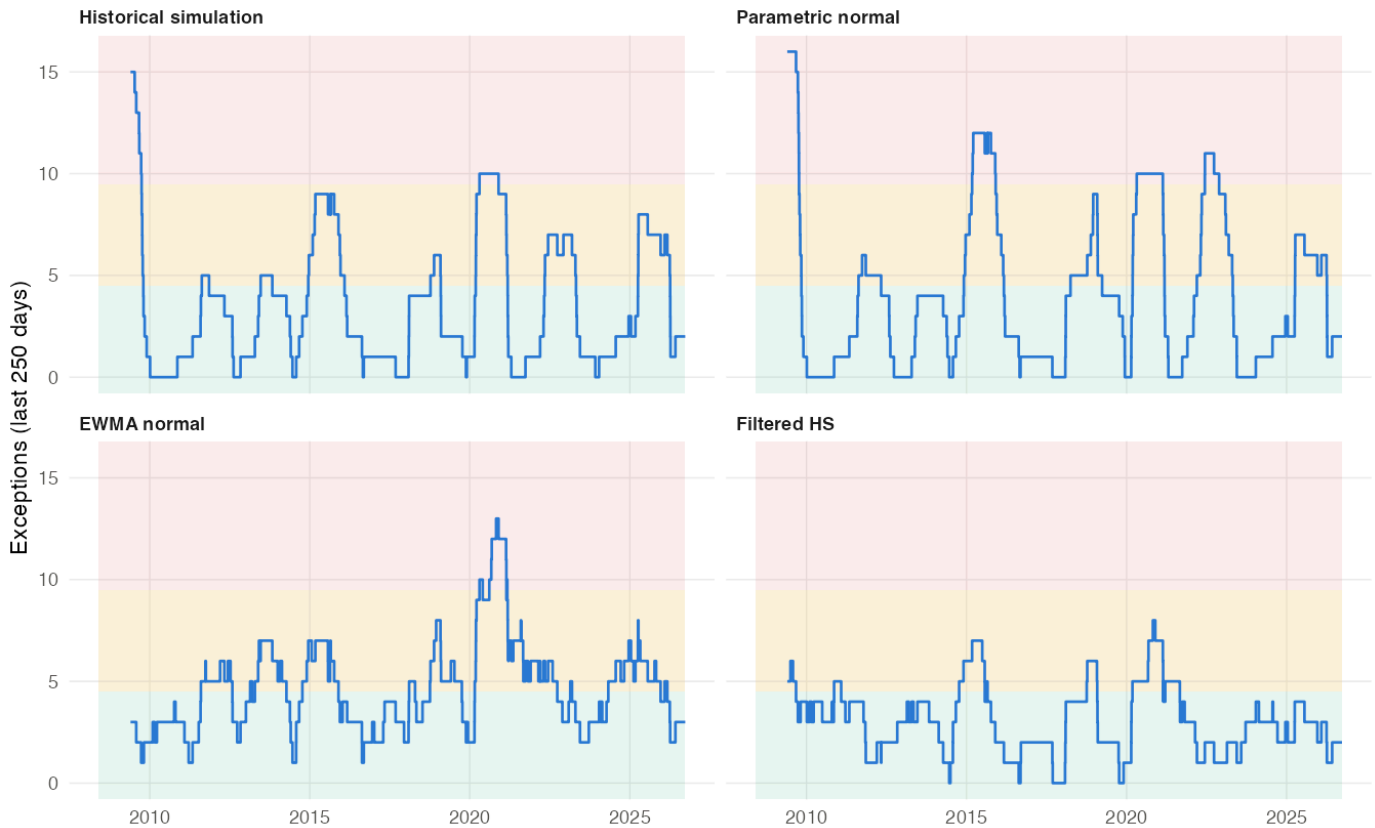
Backtest: daily P&L vs. 99% VaR forecast (made the day before)

Grey: P&L. Blue: -VaR. Red dots: exceptions (loss bigger than VaR).



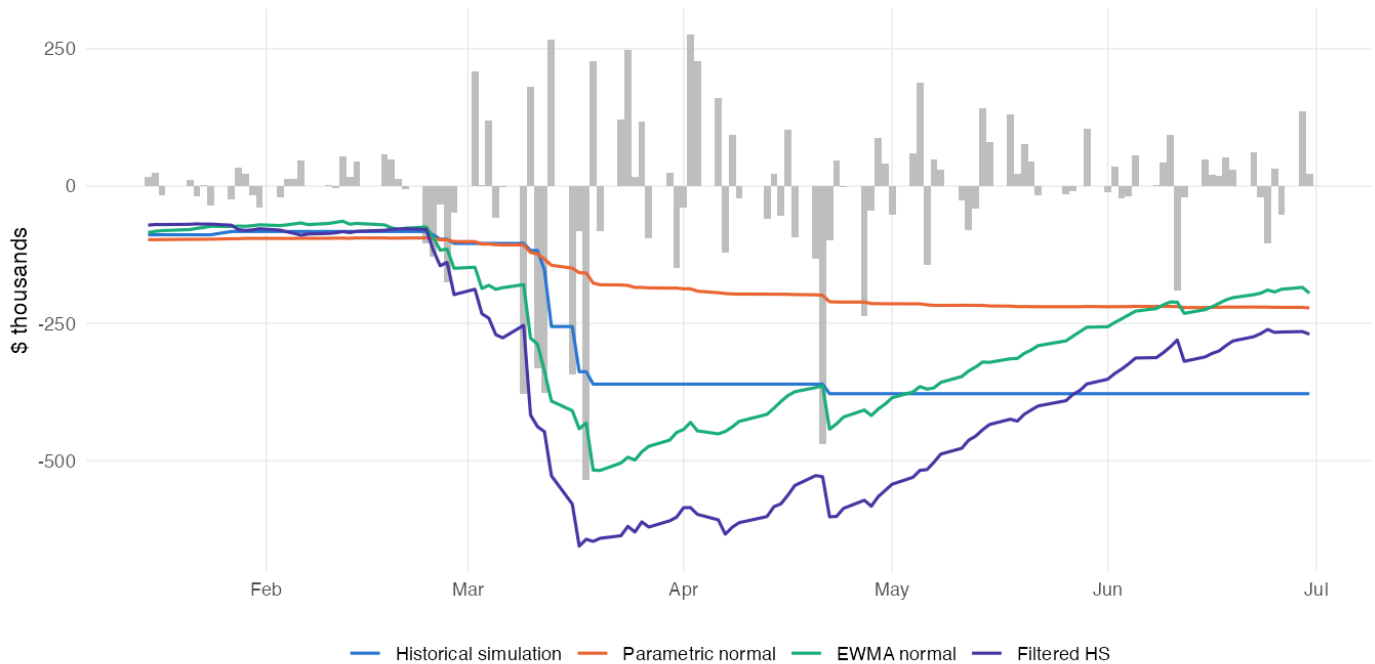
Basel traffic light: exceptions in the trailing 250 days

Green 0-4 | Yellow 5-9 (capital multiplier rises) | Red 10+ (model presumed flawed)



Reaction speed during the COVID crash (Feb-Jun 2020)

Compare how quickly each model widens its VaR after the volatility shock.

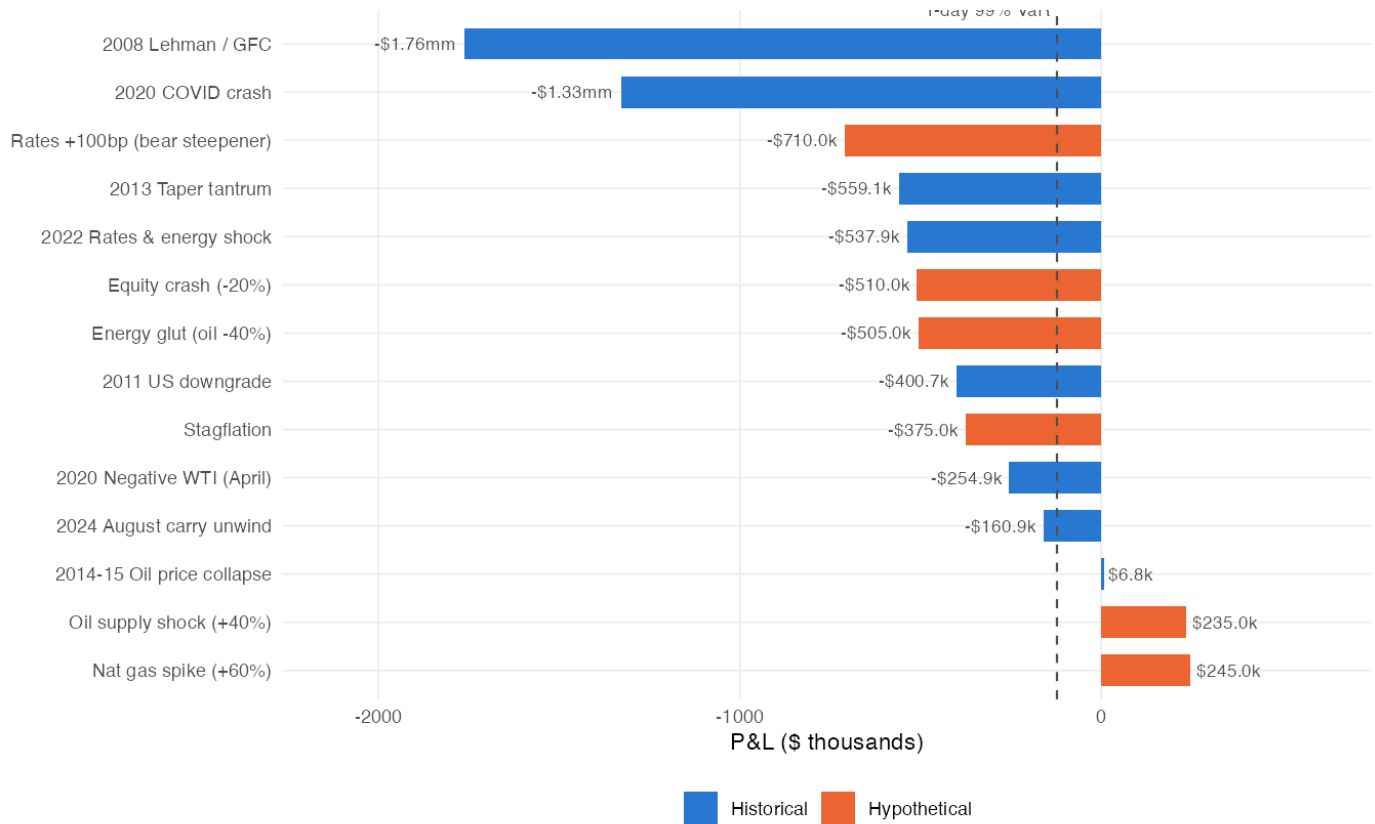


5. Stress testing

Scenario	Type	P&L	x VaR
2008 Lehman / GFC	Historical	-\$1.76mm	14.4
2020 COVID crash	Historical	-\$1.33mm	10.8
Rates +100bp (bear steepener)	Hypothetical	-\$710.0k	5.8
2013 Taper tantrum	Historical	-\$559.1k	4.6
2022 Rates & energy shock	Historical	-\$537.9k	4.4
Equity crash (-20%)	Hypothetical	-\$510.0k	4.2
Energy glut (oil -40%)	Hypothetical	-\$505.0k	4.1
2011 US downgrade	Historical	-\$400.7k	3.3
Stagflation	Hypothetical	-\$375.0k	3.1
2020 Negative WTI (April)	Historical	-\$254.9k	2.1
2024 August carry unwind	Historical	-\$160.9k	1.3
2014-15 Oil price collapse	Historical	\$6.8k	-0.1
Oil supply shock (+40%)	Hypothetical	\$235.0k	-1.9
Nat gas spike (+60%)	Hypothetical	\$245.0k	-2.0

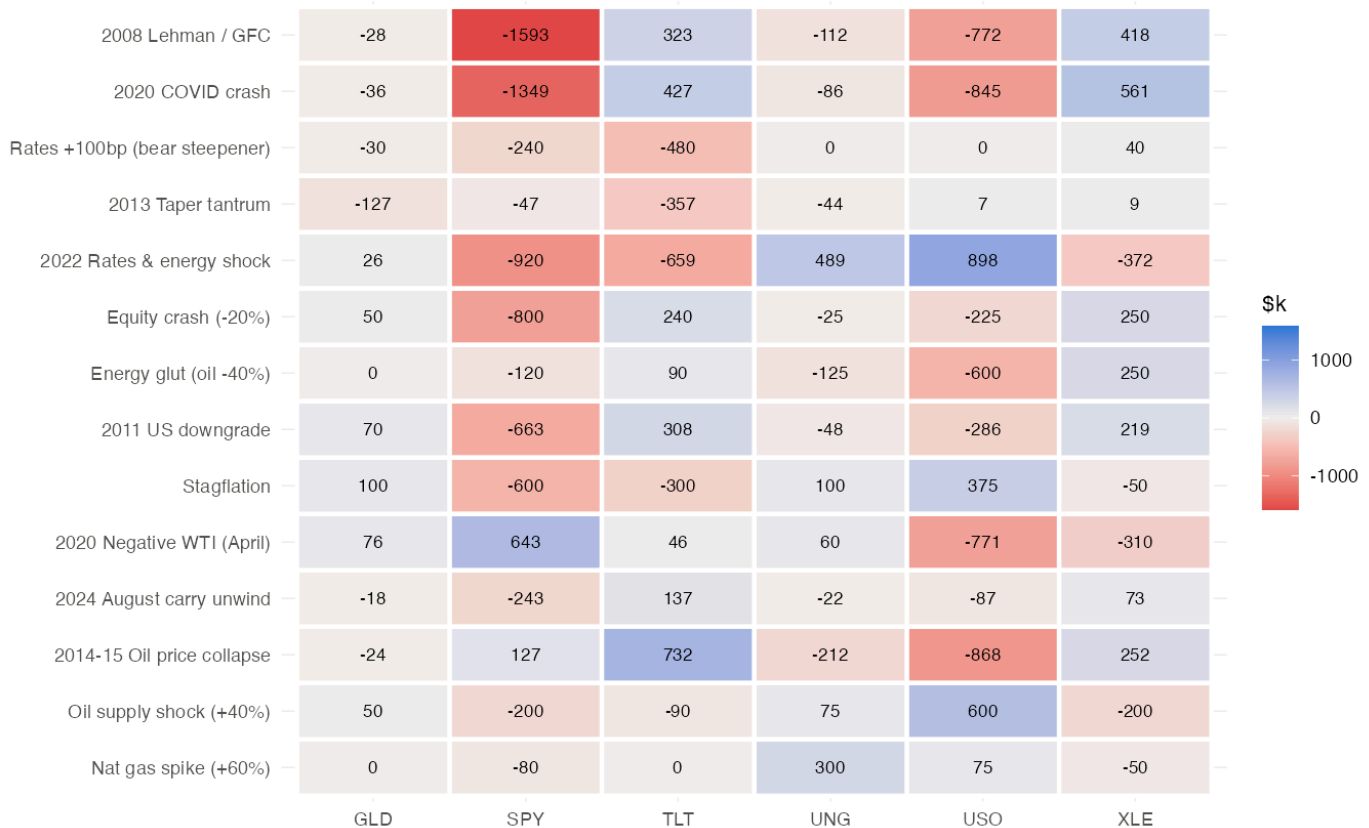
Stress test P&L on today's portfolio

Stress losses are often many multiples of VaR: VaR is not a worst case.



Stress P&L by position (\$ thousands)

Read across a row to see what hurts and what hedges in each scenario.



Regulatory notes

- **Basel 2.5 (US Market Risk Rule, in force today):** 10-day 99% VaR plus stressed VaR, each times a multiplier of 3 (up to 4 after poor backtests). Backtesting uses the traffic light on 250 days of 99% VaR exceptions.
- **FRTB (Fundamental Review of the Trading Book):** replaces VaR with 97.5% Expected Shortfall scaled to liquidity horizons (10 to 120 days), calibrated to a stressed period, and approved desk by desk. Desks must pass backtesting (at most 12 exceptions at 99% and 30 at 97.5% in 250 days) and a P&L attribution test comparing risk-model P&L to front-office P&L.
- **Timing:** FRTB capital applies in the EU from 1 Jan 2027; the UK applies Basel 3.1 from 1 Jan 2027 with FRTB internal models from 1 Jan 2028; US agencies re-proposed their Basel III endgame rules (including FRTB) in March 2026. Check current status before quoting dates.
- **Model risk (SR 11-7 in the US):** every model needs independent validation: conceptual soundness, ongoing monitoring (this backtest) and outcomes analysis.

Limitations

- ETFs proxy risk factors; a real book maps positions to curves, spreads and vol surfaces.

- Positions are linear, so there are no option Greeks (gamma, vega) in this book.
- 10-day figures use square-root-of-time scaling, which assumes independent daily returns.
- USO and UNG hold futures, so their returns include roll yield, not just spot moves.